Qns	Answer	Marks
3(a)(i)	$Q_{\text{water at 5.0°C to water at 0°C}} = m_{\text{water}} c_{\text{water}} \Delta T$ $= 220 \times 4.18 \times (5.0 - 0)$ $= 4598$ J $= 4600$ J (2 s.f.)	B1
3(a)(ii)	$Q_{\text{water at 0°C to ice at 0°C}} = m_{\text{water}} L_{\text{f}}$ $= 220 \times 334$ $= 73480$ J $= 73500$ J (3 s.f.)	B1

Qns	Answer	Marks
	Note: accept 73000 J and 73500 J.	
3(b)	$Q_{\text{ice at -8.0°C to ice at 0°C}} = m_{\text{ice}} c_{\text{ice}} \Delta T$ $= 500 \times 2.09 \times \left[0 - (-8.0)\right]$ $= 8360$ J (3 s.f.) **Note: accept 8360 J and 8400 J.**	**B1**
3(c)	0 °C Since the heat needed to raise the temperature of the original ice to 0 °C (= 8360 J) is greater than the heat released by water as its temperature decreases to 0 °C (= 4598 J), but is less than the total heat released by water as it freezes to 0 °C ice (= 4598 + 73480 = 78078 J), the original water will not fully freeze into 0 °C ice. The final state will be water-ice mixture at 0 °C. **Note:** When the temperature of the water decreases to 0 °C (releasing 4598 J of heat), the temperature of the original ice is still below 0 °C (to raise it to 0 °C, 8360 J of heat is needed). Hence, the original water will continue to partially freeze, until the original ice is at 0 °C. The final product is a water-ice mixture at 0 °C.	**A1** **M1**